

Nitish Ravisankar Raveendran

rrnitish@gmail.com

LinkedIn: Nitish Ravisankar Raveendran

GitHub: Nitish05

8801 38th Ave,
College Park, Maryland
+1 (240)-825-6896

Education

University of Maryland, College Park

Master of Engineering in Robotics

Expected Graduation: Spring, 2025

Relevant Coursework: Advanced Robotics, Machine Learning, AI in Robotics, Sensor Integration, Computer Vision, Sensor Fusion

Anna University, PEC, Chennai

Bachelor of Engineering in Electronics and Communications Engineering

Graduated: July 2022

CGPA: 8.94/10.0

Work Experience

Cognizant, Chennai

Intern

February 2022 - August 2023

- Developed features for full-stack web applications, strengthening software engineering and problem-solving skills.

Institute of Innovations, Chennai

Intern

May - June 2021

- Designed and programmed IoT devices using ESP8266 and ESP-32, enabling real-time data collection and communication.

Projects

Bipedal Locomotion and Control for Humanoid Robots

Fall 2024

- Designed closed-loop control systems for bipedal locomotion and object manipulation using ROS2.
- Implemented state estimation using IMU, encoders, and force sensors to enhance stability and precision.

Quadcopter Design and Development

Summer 2024

- Designed and built a quadcopter, integrating mechanical, electronics, and software components for autonomous flight.
- Developed custom firmware for motor control and sensor integration to achieve stable flight.
- Utilized Python and C++ for simulation and control system development.

Reinforcement Learning for Robot Pursuit and Evasion

Spring 2024

- Created a reinforcement learning framework using DQN to train two robots (evader and pursuer) in a 3D ROS2 and Gazebo environment.
- Optimized the reward system to enhance decision-making and behavioral adaptation.

Gesture-Based Control in Assistive Technology

Fall 2024

- Developed a hand-gesture control system using MediaPipe and ROS2 to teleoperate robots.
- Focused on real-time hand tracking and gesture recognition to enable intuitive robot control.
- Implemented teleoperation on TurtleBot for navigation and manipulation tasks.

Multi-Agent Navigation with Centralized and Decentralized MCTS

Fall 2024

- Compared centralized and decentralized navigation methods for multi-agent systems using Monte Carlo Tree Search (MCTS).
- Conducted extensive simulations in ROS2 and Gazebo to evaluate performance in dynamic environments.
- Contributed to modularizing the centralized system for scalability and future experiments.

Lunar Light Utility Vehicle (LLUV)

Fall 2024

- Designed a lightweight, autonomous lunar rover for site preparation and infrastructure support.
- Engineered adaptive suspension, optimized wheel designs, and integrated autonomous capabilities.
- Focused on achieving efficient navigation and payload management for extraterrestrial terrains.

Arduino-based Aimbot with YOLO Object Detection

Summer 2024

- Built a YOLO-based real-time object detection system integrated with Arduino to automate target tracking and servo alignment.

Developed Planning Methods for Manipulators and Vehicles

Spring 2024

- Designed and analyzed planning algorithms for robotic manipulators and vehicle navigation.
- Implemented and optimized methods including RRT, A*, and Potential Fields in dynamic environments.

Design and Simulation of LQR and LQG Controllers

Fall 2023

- Modeled and implemented LQR and LQG controllers for a two-mass crane system using MATLAB, analyzing control dynamics.

Alpha - Pick and Place Mobile Manipulator

Fall 2023

- Designed a CAD-based mobile manipulator with ROS2 integration for precise pick-and-place operations.

Leadership & Volunteering

Robotics Club, PEC Chennai

President

March 2019 - August 2022

National Service Scheme (NSS)

Member

November 2018 - July 2022

Technical Skills

- **Programming:** Python, C++, MATLAB
- **Machine Learning:** PyTorch, TensorFlow, Deep Reinforcement Learning
- **Robotics:** ROS2, Gazebo, Computer Vision, Perception Systems
- **Embedded Systems:** Arduino, ESP8266, ESP-32
- **Simulation and Algorithms:** Sensor Fusion, State Estimation, Depth Estimation, Path Planning
- **Tools:** JAX, Docker, Git

Certifications

- Build a Face Recognition Application using Python - GUVI, April 2021
- Python Programming - Coursera
- Advanced Robotics
- Smart India Hackathon Participant
- Web Development - Codebind

Languages & Interests

Languages: English, Tamil

Interests: Autonomous Systems, Perception Systems, UAV Design, AI Applications